5.2.12

EE25BTECH11009 - Anshu kumar ram

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Problem

Question:
Solve the system of equations

$$2x + y - 6 = 0 \tag{1}$$

$$4x - 2y + 4 = 0 \tag{2}$$

Solution Setup

The equation of line:

$$\mathbf{n}^\top \mathbf{x} = c \tag{3}$$

Line L:

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 6 \tag{4}$$

Line K:

$$\begin{pmatrix} 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -4 \tag{5}$$

Matrix Form

These can be combined and written in matrix form:

$$\begin{pmatrix} 2 & 1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \end{pmatrix} \tag{6}$$

The following augmented matrix can be solved by Gaussian elimination:

$$\left(\begin{array}{cc|c} 2 & 1 & 6 \\ 4 & -2 & -4 \end{array}\right) \xleftrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{cc|c} 2 & 1 & 6 \\ 0 & -4 & -16 \end{array}\right) \tag{7}$$

Rank and Solution

From the second row:

$$-4y = -16 \implies y = 4 \tag{8}$$

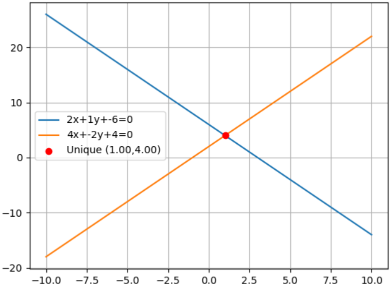
Substituting in the first equation:

$$2x + 4 = 6 \implies x = 1 \tag{9}$$

Hence, the unique solution is

$$(x, y) = (1, 4) \tag{10}$$

Figure



20
10
0
−10
−20
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
2x+1y+-6=0
4x+-2y+4=0
Unique (1.00,4.00)

C Code (code.c)

```c
#include <stdio.h>

int solve_linear(double a1, double b1, double c1,
                 double a2, double b2, double c2,
                 double *x, double *y) {
    double det = a1*b2 - a2*b1;
    if(det != 0) {
        *x = (b1*c2 - b2*c1) / det;
        *y = (c1*a2 - c2*a1) / det;
        return 1; // unique solution
    }
    if(a1*b2 == a2*b1 && a1*c2 == a2*c1 && b1*c2 == b2*c1) {
        return 2; // infinite solutions
    }
    return 0; // no solution
}
```

Python + C integration

```
import ctypes
import matplotlib.pyplot as plt
import numpy as np
# Load shared library
lib = ctypes.CDLL("./code.so")

lib.solve_linear.argtypes = [ctypes.c_double, ctypes.c_double, ctypes.
    c_double,
                                ctypes.c_double, ctypes.c_double, ctypes.
                                    c_double,
                               ctypes.POINTER(ctypes.c_double), ctypes.
                                   POINTER(ctypes.c_double)]
lib.solve_linear.restype = ctypes.c_int
```

Python + C integration

```
# New equations: 2x + y − 6 = 0, 4x − 2y + 4 = 0
a1, b1, c1 = 2, 1, −6
a2, b2, c2 = 4, −2, 4
x = ctypes.c_double()
y = ctypes.c_double()
res = lib.solve_linear(a1,b1,c1,a2,b2,c2,ctypes.byref(x),ctypes.byref(y))
# Plot
xs = np.linspace(−10, 10, 400)
y1 = −(a1*xs + c1)/b1
y2 = −(a2*xs + c2)/b2

plt.plot(xs, y1, label=f"{a1}x+{b1}y+{c1}=0")
plt.plot(xs, y2, label=f"{a2}x+{b2}y+{c2}=0")
```

Python + C integration

```
if res == 1:
    plt.scatter([x.value],[y.value], color="red", zorder=5,
                label=f"Unique ({x.value:.2f},{y.value:.2f})")
elif res == 2:
    plt.title("Infinite solutions (same line)")
else:
    plt.title("No solution (parallel lines)")
plt.legend()
plt.grid(True)
plt.show()
```

Pure python

```
# code.py
import matplotlib.pyplot as plt
import numpy as np

# New equations: 2x + y - 6 = 0, 4x - 2y + 4 = 0
a1, b1, c1 = 2, 1, -6
a2, b2, c2 = 4, -2, 4

det = a1*b2 - a2*b1
xs = np.linspace(-10, 10, 400)

if det != 0:
    x = (b1*c2 - b2*c1) / det
    y = (c1*a2 - c2*a1) / det
    res = 1
```

Pure python

```
elif (a1*b2 == a2*b1 and a1*c2 == a2*c1 and b1*c2 == b2*c1):
    res = 2
else:
    res = 0

# Plot lines
y1 = -(a1*xs + c1)/b1
y2 = -(a2*xs + c2)/b2
plt.plot(xs, y1, label=f"{a1}x+{b1}y+{c1}=0")
plt.plot(xs, y2, label=f"{a2}x+{b2}y+{c2}=0")
```

Pure python

```
if res == 1:
    plt.scatter([x],[y], color="red", zorder=5,
                label=f"Unique ({x:.2f},{y:.2f})")
elif res == 2:
    plt.title("Infinite solutions (same line)")
else:
    plt.title("No solution (parallel lines)")

plt.legend()
plt.grid(True)
plt.show()
```